



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2017 – 0486



Agricultural Machinery – Rice Combine Harvester AGRIUNION AU2.01-A

Test Report Number	2017 – 0486
Date of Issuance	February 21, 2020
Valid Until	February 21, 2025
Agricultural Machinery	Rice Combine Harvester
Type	Crawler type, Sack type, Ride-on
Brand	AGRIUNION
Model	AU2.01-A
Test Requested by	AGRI COMPONENT CORPORATION National Hi-Way, Tagaran, Cauayan City, Isabela

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 224:2015	AMTEC Test
Field Efficiency, %, minimum	75.0	50.6
Losses, %, maximum		
a) Header Loss	0.5	0
b) Separation Loss/ Blower Loss	1.3	0.21
c) Unthreshed Loss	0.2	0
d) Scattering Loss	0.3	0
Purity (With Fan), %, minimum	97	92.6
Broken Grains, %, maximum	2.0	1.46
Net Cracked Grain, %, maximum	5.0	7.67
Noise Level, dB(A), maximum	95.0*	98.9

*Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Department of Labor and Employment, Philippines. 2013.

OBSERVATIONS

1. Operator's manual (Based on PAES 102:2000)	Not provided
2. Previous Valid Test Reports of Similar Brand and Model	None
3. Manufacturer	ZOOMLION HEAVY MACHINERY ZHEJIANG CO., LTD., China
4. Prime Mover Used	53 kW Diesel Engine
5. Number of Operators	Three (One operator, two baggers)
6. Cleaning of Parts	Parts can be cleaned when not in operation.
7. Adjusting of and Repairing of Parts	Not observed
8. Transporting the Machine	Machine can be loaded to a trailer.
9. Safety Features	Warning stickers were provided on the machine.
10. Failure or Abnormalities That May Be Observed on the Thresher or Its Component Parts During and After the Threshing Operation	No breakdown or malfunction occurred during the test.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Dates of Test	May 4, 2017
Location of Test	Tumauini, Isabela
Items Tested	Field capacity, Field efficiency, Threshed grain output capacity, Grain losses, Fuel consumption rate, Noise level, and Number of operators

METHODS OF TEST

The rice combine harvester was tested based on the PNS/PAES 225:2015 Agricultural Machinery – Rice Combine Harvester – Methods of Test.

DESCRIPTION OF THE MACHINE

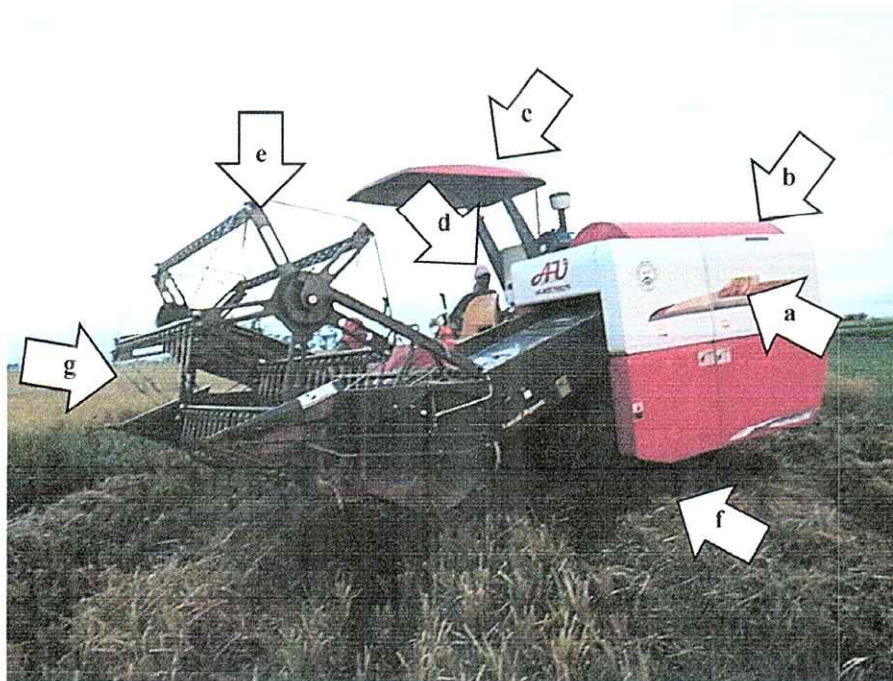


Figure 1. Parts of the AGRIUNION AU2.01-A Rice Combine Harvester powered by 53 kW Diesel Engine include the following: (a) Threshing unit, (b) Grain bagging tank, (c) Canopy, (d) Operator's seat, (e) Reel assembly, (f) Track-type traveling device, and (g) Tines.

SPECIFICATIONS

Item	Manufacturer's Specifications	AMTEC Verification
1 Overall dimensions and weight of rice combine harvester		
1.1 Length, mm	4820	4875
1.2 Width, mm	2560	2700
1.3 Height, mm	2660	2420
1.4 Total weight of the machine, kg	2800	NM
2 Crop conditions		
2.1 Variety	ND	Front line 12 Hybrid
2.2 Maximum cutting angle of straw, °	ND	NM
2.3 Plant height, mm	ND	629
2.4 Crop moisture content, %	ND	28.9
3 Harvesting conditions		
3.1 Number of cutting rows	ND	ND
3.2 Cutting method	ND	ND
3.3 Cutting width, mm	ND	2060*
3.4 Minimum cutting height, mm	ND	354
3.5 Field capacity, ha/h		
4 Starting system	ND	Electric start
5 Operating condition	ND	ND
6 Harvesting speed, m/s	ND	1.42*
7 Ground drive	ND	Crawler

*Actual measurement during the performance test

Note:– No Data, NM – Not Measured

RESULTS

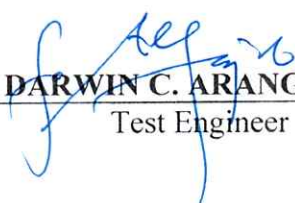
Item		Data
1	Crop conditions	
1.1	Variety	Front line 12 (Hybrid)
1.2	Number of days after sowing	ND
1.3	Plant spacing, mm	20 x 20
1.4	Recommended period of maturity	ND
1.5	Average plant height, mm	629
1.6	Lodging angle of plant, °	0
1.7	Plant population/m ²	36
1.8	Number of tillers/plant	NM
1.9	Grain-straw ratio	57
1.10	Grain moisture content, % _{owb}	28.9
2	Field conditions	
2.1	Location of test field	Tumauini, Isabela
2.2	General topography	Leveled
2.3	Area of test field, m ²	651
2.3.1	Length, m	10.5
2.3.2	Width, m	62
2.4	Shape of field	Rectangular
2.5	Soil type	ND
2.6	Soil hardness, kg/cm ²	2.74
3	Performance test	
3.1	Date of test	May 4, 2017
3.2	Duration of test, h	0.096
3.3	Time lost, h	0.038
3.4	Operating speed, m/s	1.42
3.5	Effective width of cut, mm	1975

Note: ND – No Data, NM – Not Measured

RESULTS (Cont'n)

Item		Data
3.6	Fuel consumption rate, L/h	10.1
3.7	Specific fuel consumption, L/ha	15.4
3.8	Noise level, dB(A)	98.9
3.9	Height of cut or height of stubbles, mm	354
3.10	Threshed grain output capacity, kg/h	3579.1
3.11	Actual field capacity	
3.11.1	ha/h	0.679
3.11.2	h/ha	1.472
3.12	Theoretical field capacity	
3.12.1	ha/h	1.340
3.12.2	h/ha	0.745
3.13	Field efficiency, %	50.6
4	Laboratory grain analysis	
4.1	Purity, %	92.6
4.2	Total losses, %	0.21
4.2.1	Header loss, %	0
4.2.2	Blower/Separation loss, %	0.21
4.2.3	Unthreshed loss, %	0
4.2.4	Scattering loss, %	0
4.3	Potential yield, kg/m ²	1.20
4.4	Broken grains, %	1.46
4.5	Net cracked grains, %	7.67

Tested by:


DARWIN C. ARANGUREN

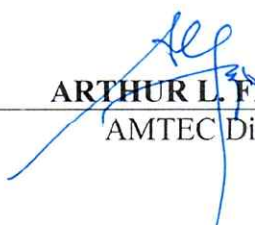
Test Engineer

Verified by:

ROMULO E. EUSEBIO

Test Engineer

Approved for Release:


ARTHUR L. FAJARDO

AMTEC Director

FEB 24 2020

Date

REMINDER

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